

# The phenylalanine ammonia-lyase gene family in *Salvia miltiorrhiza*: genome-wide characterization, molecular cloning and expression analysis

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**Abstract** *Salvia miltiorrhiza* Bunge is a well-known material of traditional Chinese medicine. Hydrophilic phenolic acids, such as rosmarinic acid and salvianolic acid B, are a group of pharmaceutically important compounds in *S. miltiorrhiza*. The biosynthesis of rosmarinic acid requires the coordination of the phenylpropanoid pathway and the tyrosine-derived pathway. Phenylalanine ammonia-lyase (PAL) is the first key enzyme of the phenylpropanoid pathway. Systematic analysis of the *SmPAL* gene family has not been carried out. We report here the identification of three *SmPALs* through searching the recently obtained working draft of the *S. miltiorrhiza* genome and full-length cDNA cloning. Bioinformatic and phylogenetic analyses showed that *SmPAL1* and *SmPAL3* clustered in a subclade of dicot PALs, whereas *SmPAL2* fell into the other one. Some important *cis*-elements were conserved in three *SmPAL* promoters, whereas the others were not. *SmPAL1* and *SmPAL3* were highly expressed in roots and leaves of *S. miltiorrhiza*, but *SmPAL2* were predominately expressed in stems and flowers. It indicates that *SmPAL1* and *SmPAL3* function redundantly in rosmarinic acid biosynthesis. All *SmPALs* were induced in roots treated with PEG and MeJA, but the time and degree of responses were different, suggesting the complexity of *SmPAL*-associated metabolic network in *S. miltiorrhiza*. This is the first comprehensive study dedicated to *SmPAL* gene family

characterization. The results provide a basis for elucidating the role of *SmPAL* genes in the biosynthesis of bioactive compounds.

**Keywords** Phenolic acid · Phenylalanine ammonia-lyase · Rosmarinic acid · Salianolic acid B · *Salvia miltiorrhiza* · Traditional Chinese medicine

## Abbreviations

|         |                                                  |
|---------|--------------------------------------------------|
| 4CL     | 4-Coumarate:CoA ligase                           |
| C4H     | Cinnamic acid 4-hydroxylase                      |
| HPPD    | Hydroxyphenylpyruvate dioxygenase                |
| HPPR    | Hydroxyphenylpyruvate reductase                  |
| MeJA    | Methyl jasmonate                                 |
| MW      | Molecular weight                                 |
| ORF     | Open reading frame                               |
| PAL     | Phenylalanine ammonia-lyase                      |
| PCR     | Polymerase chain reactions                       |
| pI      | Isoelectric point                                |
| qRT-PCR | Quantitative real-time polymerase chain reaction |
| RACE    | Rapid amplification of cDNA ends                 |
| RAS     | Rosmarinic acid synthase                         |
| TAT     | Tyrosine aminotransferase                        |
| TCM     | Traditional Chinese medicine                     |

## Introduction

*Salvia miltiorrhiza* Bunge is a well-known material of various traditional Chinese medicines (TCMs) that have been widely used for treating dysmenorrhoea, amenorrhoea, and cardiovascular diseases [1]. The active pharmaceutical ingredients of *S. miltiorrhiza* mainly include lipophilic tanshinones and hydrophilic phenolic acids. The

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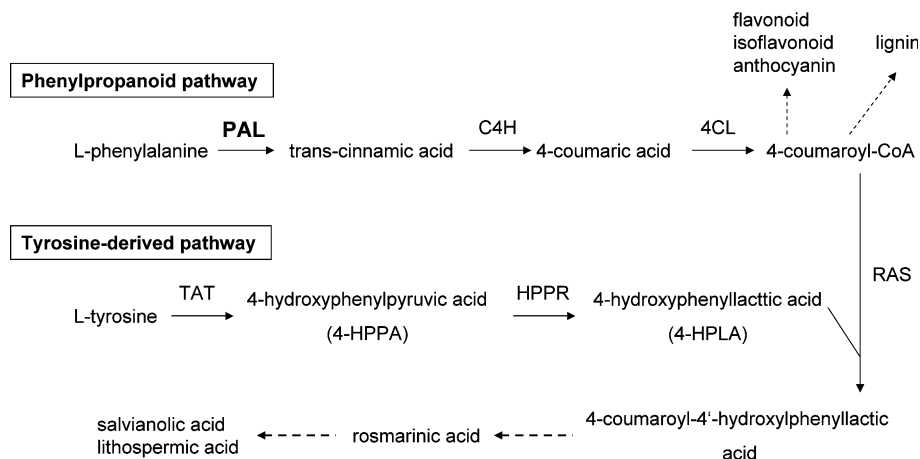
latter is a large group of chemicals, including rosmarinic acid, salvianolic acid A, salvianolic acid B, lithospermic acid, and so forth [2]. Among them, salvianolic acid B is the major component of active phenolic acids in *S. miltiorrhiza*, while rosmarinic acid, an ester of 3,4-dihydroxyphenyllactic acid and caffeic acid, serves as a basis of salvianolic and lithospermic acids. The biosynthesis of rosmarinic acid requires the coordination of two independent pathways, the phenylpropanoid pathway and the tyrosine-derived pathway. The phenylpropanoid pathway is responsible for the biosynthesis of caffeic acid, whereas the tyrosine-derived pathway is devoted to 3,4-dihydroxyphenyllactic acid biosynthesis (Fig. 1). Key enzymes involved in the two pathways include at least phenylalanine ammonia-lyase (PAL), cinnamic acid 4-hydroxylase (C4H), 4-coumarate:CoA ligase (4CL), tyrosine aminotransferase (TAT), hydroxyphenylpyruvate reductase (HPPR), hydroxyphenylpyruvate dioxygenase (HPPD), and rosmarinic acid synthase (RAS) (Fig. 1).

Phenylalanine ammonia-lyase (EC 4.3.1.5) is the first key enzyme in the phenylpropanoid pathway. It catalyzes the deamination of L-phenylalanine to produce trans-cinnamic acid, which is then converted to *p*-coumaroyl-CoA by C4H and 4CL for the production of lignin, flavonoid, coumarin, isoflavonoid, furanocoumarin, norlignan, salvianolic acid B, and many other secondary metabolites. PAL was first purified from *Hordeum vulgare* by Koukol and Connth [3], but was shown to widely exist in plants, fungi, viruses and algae afterwards. So far, there are no reports for animal PALs. However, it has been shown that PAL can play significant roles in animals and has commercial and medical potential. PAL is able to substantially inhibit neoplastic cell growth in vitro [4] and has the potential to treat the inherited metabolic disorder [5]. Additionally, PAL may be used in large-scale bio-conversion of trans-cinnamic acid and ammonium salts into L-phenylalanine, since the reaction catalyzed by PAL is reversible [6].

Phenylalanine ammonia-lyase is encoded by a small multigene family in most plant species. There are four *PAL* genes in *Arabidopsis thaliana* [7, 8], five in *Populus trichocarpa* [9, 10], three in *Scutellaria baicalensis* [11], seven in *Cucumis sativus* [12], and three in *Coffea canephora* [13]. The members of *PAL* gene family in a plant are usually expressed differentially in tissues and in response to environmental stimuli and appear to be functionally distinct. For instance, among three *C. canephora* *PALs*, *CcPAL1* showed high expression in roots, small green-stage beans and pericarps, followed by flowers, red-stage pericarps, branches, large green-stage beans, and less in other tissues analyzed. *CcPAL2* showed the highest expression in flowers, followed by pericarps and branches, and less in beans, roots and leaves. Whereas, *CcPAL3* was predominantly expressed in small green-stage beans and pericarps, large green- and red-stage pericarps, very low expression in other tissues [13]. Similarly, seven *C. sativus* *PAL* genes were also differentially expressed in roots, stems, cotyledons, leaves, flowers and fruits [12]. Analyzing the functions of *P. tremuloides* *PALs* showed that *PtPAL1* could be involved in the biosynthesis of condensed tannins and other phenolics, whereas *PtPAL2* was most likely to be responsible for lignin biosynthesis [14]. However, systematic analysis of the *PAL* gene family has not been carried out for *S. miltiorrhiza*.

Recently, a working draft of the *S. miltiorrhiza* genome has been obtained (Chen et al. unpublished data). In order to get a clear picture of the *SmPAL* gene family, we searched the current assembly of *S. miltiorrhiza* genome and then performed PCR amplification of full-length cDNAs. A total of three *SmPALs* were identified. Gene structures and sequence characteristics of *SmPALs* were subsequently analyzed. Expression level of *SmPAL* genes in various tissues of *S. miltiorrhiza* with or without stress treatment was examined by quantitative real-time RT-PCR. The results suggest the complexity of *SmPAL*-associated metabolic network and the functional redundancy of *SmPAL1*

**Fig. 1** Proposed pathways of rosmarinic acid biosynthesis in plants



and *SmPAL3* in rosmarinic acid biosynthesis. It provides a basis for elucidating the role of *SmPAL* genes in the biosynthesis of secondary metabolites, particularly the bioactive rosmarinic acid and salvianolic acid B.

## Materials and methods

### Plant materials and stress treatment

*Salvia miltiorrhiza* Bunge (line 993), whose genome has been sequenced, was grown in a field nursery. Roots, stems, leaves and flowers were collected from 2-year-old plants in August. Methyl jasmonate (MeJA) treatment was carried out as described previously [15]. Drought treatment was performed by adding 15 % PEG-6000 to 6,7-V liquid media with plantlets pre-cultivated for 2 days. Plantlets were treated for 12, 24, 36 and 48 h and then roots were sampled. Plantlets treated with sterile water were used as controls. The treatments were repeated three times. All samples were frozen immediately after collected and stored in liquid nitrogen until use.

### Identification of genomic sequence of *SmPAL* genes

The genomic sequence of *SmPALs* were identified by BLAST analysis of four known *Arabidopsis* AtPALs (NM\_129260, NM\_115186, NM\_120505, NM\_111869) against the current assembly of *S. miltiorrhiza* genome (Chen et al. unpublished data) using the BLASTx algorithm [16]. An e-value cut-off of  $10^{-5}$  was applied to the homologue recognition. Gene models were predicted as described previously [15].

### Cloning of the full-length *SmPAL3* cDNA

Total RNA was extracted from roots of *S. miltiorrhiza* using the general plant total RNA extraction kit (Bioteke, Beijing, China). Poly(A)<sup>+</sup> RNA was isolated from total RNA using the Oligotex mRNA purification kit (QIAGEN, MD, CA, USA). Rapid amplification of cDNA ends (RACE) were performed using the GeneRacer™ kit (Invitrogen, Carlsbad, CA, USA). PCR amplification of 5' ends was carried out using the GeneRacer 5' primer and the nesting gene-specific primer PAL3R1 (5'-CCTGAACCTCC TCCACCATCCTCTTCA-3'). Nested PCR amplification of 5' ends was performed using the GeneRacer 5' nested primer and the nested gene-specific nested primer PAL3R2 (5'-CTCTGCTTCACGCAGAACCCGTTCT-3'). PCR amplification of 3' ends was carried out using the GeneRacer 3' primer and the nesting gene-specific primer PAL3F1 (5'-GAGCAATGGTTTGCTCGTTGATCCT-3'). Nested PCR amplification of 3' ends was performed using

the GeneRacer 3' nested primer and the nested gene-specific primer PAL3F2 (5'-GGTTTGCTCGTTGATCCTT TGTGA-3'). PCRs were performed in a kit-recommended 25 µL standard amplification system using 1.5 µL of total first strand cDNA as a template for primary amplification and 0.5 µL of the primary amplification products as a template for nested amplification. The primary amplification reactions for both 5' and 3' ends were carried out under the following conditions: predenaturation at 94 °C for 2 min, 5 cycles of amplification at 94 °C for 30 s and 72 °C for 1 min, 5 cycles of amplification at 94 °C for 30 s and 70 °C for 1 min, 25 cycles at 94 °C for 30 s, 60 °C for 45 s and 72 °C for 1.5 min, followed by a final extension at 72 °C for 15 min. The nested amplification reactions for both 5' and 3' ends were carried out under the following conditions: predenaturation at 94 °C for 2 min, 30 cycles at 94 °C for 30 s, 60 °C for 45 s and 72 °C for 1.5 min, followed by a final extension at 72 °C for 15 min. PCR products were gel-purified, cloned and then sequenced.

Based on the obtained 5' and 3' cDNA sequence, a pair of gene-specific primers, including the forward primer FPAL3 (5'-GTGTGAGCGACTTTCTCTCTCATCT-3') and the reverse primer RPAL3 (5'-GCGGCTCTCCA TTCCACGATTCA-3'), were designed for the amplification of full-length *SmPAL3* cDNA. PCR was carried out in a 50 µL volume containing 1 µL cDNA template, 1 µL 10 µM FPAL3, 1 µL 10 µM RPAL3, and 25 µL Premix LA Taq. PCR conditions included 94 °C for 2 min, 35 cycles of 94 °C for 30 s, 58 °C for 45 s and 72 °C for 2.5 min, and 72 °C for 15 min. PCR products were gel-purified, cloned and then sequenced.

### Cloning of the 5' region of *SmPAL2* cDNA

5'-RACE was performed using the GeneRacer™ kit (Invitrogen) as described above for *SmPAL3*. cDNA used for primary amplification was transcribed from mRNA isolated from stems of *S. miltiorrhiza*. The nesting and nested gene-specific primers used in the reactions were PAL2R1 (5'-CCTTGAGCAATGTGTTGATTCTT-3') and PAL2R2 (5'-GCAGGGTGTGGTACGAATCACTAT-3'), respectively.

### Bioinformatic analysis and phylogenetic tree construction

The molecular weight (MW) and theoretical isoelectric point (pI) were predicted using the Compute pI/MW tool on the ExPASy server [17] ([http://web.expasy.org/compute\\_pi/](http://web.expasy.org/compute_pi/)). Known PAL sequences shown high identities with the deduced amino acid sequence of SmPALs were identified by BLASTP analyses of SmPALs against the database of non-redundant protein sequences in NCBI using the default

alignment parameters (<http://blast.ncbi.nlm.nih.gov/Blast.cgi>). The alignment of deduced amino acid sequences of SmPALs and PALs from other organisms was performed using ClustalW version 1.83 using the default parameters [18]. Phylogenetic tree was constructed using MEGA version 4.0 [19]. The reliability of branching was assessed by the bootstrap re-sampling method using 1,000 bootstrap replicates and only nodes supported by bootstrap values >50 % are shown.

### Quantitative real-time PCR

Total RNA was isolated from the tissues of *S. miltiorrhiza* with or without stress treatment using the general plant total RNA extraction kit (Bioteke, Beijing, China) and then digested with RNase-free DNase I to remove the genomic DNA contamination. Reverse transcription was carried out by SuperScript III Reverse Transcriptase. The reaction was achieved by incubating at 65 °C for 5 min, 50 °C for 1 h, and 70 °C for 15 min as described [20]. The resulting cDNA was diluted to 500 µL with sterile water. Quantitative Real-time PCR was carried out in triplicate reactions using the BIO-RAD CFX system. Gene-specific primers used were GATCTCTTCACGGAAGACCGTTGAA and GCCATTGACGCCCATTTGTGAGAGTT for *SmPAL1*, CAGGATCAAGGGGAGCAGATCGTA and GCACCATTCATTCCTGAGACA for *SmPAL2*, and CCCGCGATCGGGAA-CAGGATCAA and GCGGCTCTCCATTCCACGATTCA for *SmPAL3*. Ubiquitin gene was used as a reference as described previously [15]. The forward and reverse primers for amplification of ubiquitin were AGATGGGCGGACACTTGCTGATTA and ACTCTCCACCTCCAAAGTGATGGT, respectively. PCRs were carried out in a 20 µL volume containing 2 µL diluted cDNA, 10 µM forward primer, 10 µM reverse primer and 1 × SYBR Premix Ex TaqII using the following conditions: 95 °C for 30 s, 39 cycles of 95 °C for 5 s, 60 °C for 18 s and 72 °C for 15 s. Relative abundance of transcripts was determined using the comparative Cq method [21]. For analyzing the responses of *SmPALs* to MeJA and PEG-6000 treatments, the expression of *SmPALs* in roots without treatment was used as controls and was arbitrarily set to 1. Standard deviations were calculated from three PCR replicates.

## Results

### Genome-wide identification and molecular cloning of *SmPALs*

The decoding of the *S. miltiorrhiza* genome (Chen et al. unpublished data) enables us to perform a genome-wide search of *SmPALs* through BLAST analysis of four known

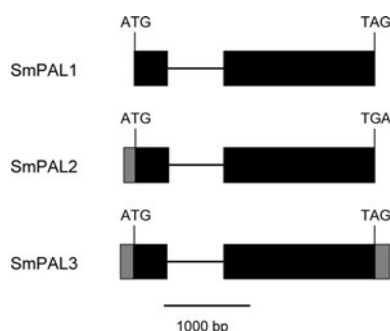
*Arabidopsis* AtPALs (NM\_129260, NM\_115186, NM\_120505, NM\_111869) against the current assembly of *S. miltiorrhiza* genome using the BLASTx algorithm [16]. As a result, we identified a total of three genomic loci of *PAL* genes. It includes the known *SmPAL1* (accession nos. EF462460 for genomic DNA and DQ408636 for cDNA) [22–24], the genomic sequence of *SmPAL2* with only partial cDNA sequence available (accession no. GQ249111) and the third *PAL* gene, *SmPAL3*, which has never been reported. We then cloned the full-length *SmPAL3* cDNA and the 5' region of *SmPAL2* cDNA using PCR technology. Taken together, it suggests that both the genomic sequence and the cDNA sequence of three *SmPALs* have been obtained and are available for comparative analysis.

Analysis of cDNA sequence of *SmPALs* showed that the open reading frames (ORFs) of *SmPAL1*, *SmPAL2* and *SmPAL3* are 2,133, 2,127, and 2,127 bp, respectively, suggesting *SmPALs* are highly similar in the length of ORF regions (Table 1). However, substantial difference was observed for sequence identity among them. *SmPAL1* and *SmPAL3* shares 85 % sequence identity, whereas the identity between *SmPAL1* and *SmPAL2* and between *SmPAL2* and *SmPAL3* is only 75 and 77 %, respectively, suggesting *SmPAL2* has less homology to *SmPAL1* and *SmPAL3* than that between *SmPAL1* and *SmPAL3*. Comparison analysis of genomic sequence and cDNA sequence showed that all of three *SmPAL* genes contain only one intron and the exon-intron junctions satisfy the GT-AG rule for donor/acceptor sites (Fig. 2). The intron is located at 386, 380 and 380 bp from ATG of *SmPAL1*, *SmPAL2*, and *SmPAL3*, respectively. The first exon of *SmPAL1* includes 387 bp coding sequence, whereas the coding sequence in the first exon of both *SmPAL2* and *SmPAL3* is 381 bp in length. The length of coding sequence in the second exon (1,750 bp) is conserved among three *SmPAL* genes. The predicted MW and theoretical pI of *SmPAL1*, *SmPAL2* and *SmPAL3* proteins are 77.1 kDa and 6.06, 77.2 kDa and 5.98, 76.5 kDa and 5.96, respectively (Table 1), which are consistent with previous results showing the MW of plant PALs varied from 72 to 83 kDa [25].

The deduced amino acid sequence of *SmPAL1*, *SmPAL2* and *SmPAL3* includes 711, 709, and 709 amino acids, respectively, and shares 80 % identity between *SmPAL1* and *SmPAL2*, 79 % between *SmPAL2* and *SmPAL3*, and 90 % between *SmPAL1* and *SmPAL3* (Fig. 3). It suggests that *SmPAL1* and *SmPAL3* are highly homologous, whereas *SmPAL2* has relatively low homology with both *SmPAL1* and *SmPAL3*, which is consistent with the results from cDNA sequence analysis of *SmPALs*. Additionally, *SmPALs* share over 80 % identity with many PALs from other plant species, such as *Melissa officinalis* MoPAL [26], *Agastache rugosa* ArPAL [27], *Perilla*

**Table 1** Properties of *SmPAL* genes and their deduced amino acid sequences

| Gene          | ORF (bp) | Protein     |          |           |
|---------------|----------|-------------|----------|-----------|
|               |          | Length (aa) | MW (kDa) | <i>pI</i> |
| <i>SmPAL1</i> | 2,133    | 711         | 77.1     | 6.06      |
| <i>SmPAL2</i> | 2,127    | 709         | 77.2     | 5.98      |
| <i>SmPAL3</i> | 2,127    | 709         | 76.5     | 5.96      |

**Fig. 2** *SmPAL* gene structures. Filled boxes represent exons with coding regions in black and 5'- and 3'-UTRs in grey. The connecting lines represent introns. Start and stop codes are shown

*frutescens* PfPAL, *S. baicalensis* SbPAL1-3 [11], *S. viscidula* SvPAL [28], and *A. thaliana* AtPAL1-4 [7, 8], confirming the role of the identified *SmPALs* in encoding PAL in *S. miltiorrhiza*. In order to further characterize *SmPALs*, we constructed a phylogenetic tree of three *SmPALs* and 57 PALs from other organisms. As shown in Fig. 4, PALs from fungus, gymnosperms, monocots and dicots clustered to four clades and PALs from dicots might be divided into four sub-clades. *SmPAL1*, *SmPAL2* and *SmPAL3* were most closely related to ArPAL, SbPAL1, and SvPAL, respectively. Like *S. miltiorrhiza*, *A. rugosa* and *S. baicalensis* and *S. viscidula* are members of the Lamiaceae family and are perennial herbs. *SmPAL1* and *SmPAL3* clustered to a sub-clade with rosmarinic acid biosynthesis-related MoPAL [26], *Cistanche deserticola* CdPAL associated with the biosynthesis of phenolic compounds [29], *Rehmannia glutinosa* RgPAL responsive to oxidative stress [30], and various PALs with unknown functions, whereas *SmPAL2* clustered to the other sub-clade with several function-unknown PALs (Fig. 4). It suggests the sequence diversity of *SmPAL2* from *SmPAL1* and *SmPAL3*.

#### Comparative analysis of the 5'-flanking regions of *SmPAL* genes

In order to investigate the transcriptional activity of three *SmPAL* gene promoters, 750 bp 5'-flanking sequence of the coding region was extracted from each *SmPAL* gene and analyzed using the PLACE (<http://www.dna.Affrc.go.jp/>)

PLACE/) and PlantCARE databases (<http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>) to identify putative *cis*-elements [31, 32]. The results showed that there were several *cis*-elements conserved in the promoters of three *SmPALs*. It includes the box L (YCYACCWACC), AC element (CYCACCWACC), MYB2 recognition site (YA-ACKG), E-box (CANNTG), and so on (Table 2). Box L is one of the three motifs (box P, YTYMMCMAMCMMC; box A, CCGTCC; and box L) previously shown to exist in plant PAL, C4H and 4CL gene promoters [33]. Box L is tightly co-regulated with box A and box P and appear to be necessary but not sufficient for elicitor- or light-mediated gene activation in parsley [33]. However, in *S. miltiorrhiza*, box A exist only in the promoter of *SmPAL3* and no box P was found in all *SmPAL* gene promoters (Table 2). The absence of one or two of box L and box A and box P was also observed in the promoters of *P. trichocarpa* and *P. tremuloides* C4H genes with distinct physiological functions [34]. It indicates that the different combination of these boxes may result in a remarkable degree of variability in gene functions. The sequences of AC element and box L are overlapped and appear to play similar roles. MYB2 recognition site (YAACKG) is a *cis*-acting element associated with drought-induced gene expression [35]. The presence of this element in all of the *SmPAL* gene promoters indicates *SmPALs* to be drought-induced. E-box is a *cis*-element recently found to be responsible for inducing gene expression in response to MeJA treatment [36]. Examination of the promoters of *SmPALs* revealed that there are four E-box *cis*-elements in *SmPAL2* promoter, whereas the number of E-box *cis*-element in *SmPAL1* and *SmPAL3* was only one (Table 2). It suggests that all three *SmPALs* are probably induced by MeJA treatment to various degrees.

In addition to the elements conserved among three *SmPALs*, many elements were found only in the promoters of one or two *SmPALs*. For instance, AG motif (AGATCCAA), a *cis*-element sufficient to confer wounding and elicitor responsiveness [37], exists only in the promoter of *SmPAL1* (Table 2). These results indicate that *SmPALs* may commonly response to some biotic and abiotic stresses, whereas in the other cases, the response can be differential for individual *SmPAL*. The underlying mechanisms of *SmPALs* in response to stress are complicated and need to be further investigated.

#### Tissue-specific expression of *SmPAL* genes

The expression levels of *SmPAL1*, *SmPAL2* and *SmPAL3* in various *S. miltiorrhiza* tissues, including roots, stems, leaves and flowers, were analyzed using qRT-PCR. The transcripts of all three *SmPAL* genes could be detected in all of the tissues analyzed, but showed differential



**Fig. 3** Alignment of the deduced SmPAL amino acid sequences

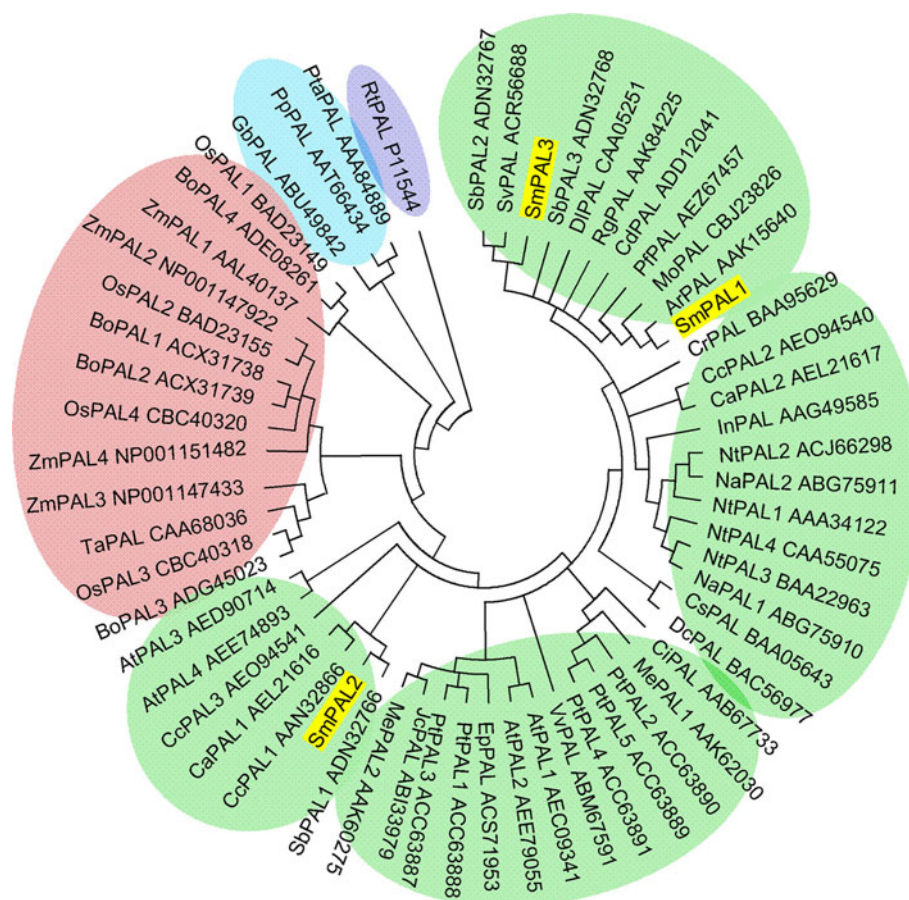
|        |                                                               |     |
|--------|---------------------------------------------------------------|-----|
| SmPAL1 | MAAENGHHEESNGFCVKQNDPLNWVAAAEELKGSHLDEVKRMVEEFRKPVVKLGGETLTI  | 60  |
| SmPAL2 | ..MENAFESSLCVESIRSEDPMLMKAASLAGSHLEEVKRMVGQYRKGVVRLGGETLTI    | 58  |
| SmPAL3 | MAAESGHSE..NGFCVKQSDPLNWAAAAEALQGSHLDAVKRMVEEFRKPAVKLGGETLTI  | 58  |
| SmPAL1 | SQVAAIAAKDNAVAVELVESSRAGVKASSDWVMSMSKGTDSYGVTTFGATSHRRTKQG    | 120 |
| SmPAL2 | GQVAAVAGREGEVKVELAAEAAREGVAASSDWVMSMNKGTDSYGVTNGFATSHRRTKQG   | 118 |
| SmPAL3 | AQVAAIASRDNAVAVELAESARAGVKASSDWVMSMGKGTDSYGVTTFGATSHRRTKQG    | 118 |
| SmPAL1 | GALQKELIRFLNAGIFGNGTESNHTLPHATRAAMLVRINTLLQGYSGIRFEILEAITKF   | 180 |
| SmPAL2 | GALQKELIRFLNAGIFGNDSDSYHTLPRSTTRASMLVRINTLLQGYSGIRFEILEIAIAKL | 178 |
| SmPAL3 | GALQKELIRFLNAGIFGNGTESNHTLPHSATRAAMLVRINTLMQGYSGIRFEILEALAKF  | 178 |
| SmPAL1 | LNENITPCLPLRGITITASGDLVPLSYIAGLLTGRPNKAVGPNGEPLNAEEAFKLAGVKG  | 240 |
| SmPAL2 | INTGITPCLPLRGITITASGDLVPLSYIAALLGRPNKAIIGPNRELLDAGRAFLAGIAT   | 238 |
| SmPAL3 | LNHNVTPLCLPLRGITITASGDLVPLSYIAGLLTGRPNKAVGPAGEPLTAEAAFKLAGVAG | 238 |
| SmPAL1 | GFFELQPK EGLALVNGTAVGSGLASIALFDANILAVLSEVMSAVFAEVMNGKPEFTDHLT | 300 |
| SmPAL2 | GFFDLQPK EGLALVNGTAVGSGLASTVLFANILAVLSEVASAVFAEVMGKPEFTDHLT   | 298 |
| SmPAL3 | GFFELQPK EGLALVNGTAVGSGLASIALYEANILALLAEVMSAIFAEVMNGKPEFTDHLT | 298 |
| SmPAL1 | HKLKHHPGQIEAAAIMEHILDGSGYVKAQKLHEQDPLQKPKQDRYALRTSPQWLGPQIE   | 360 |
| SmPAL2 | HKLKHHPGQIEAAAIMEHILDGSSYIKEAQRVHEMDPLQKPKQDRYALRTSPQWLGPQIE  | 358 |
| SmPAL3 | HKLKHHPGQIEAAAIMEHILDGSAYVKAQKLHEMDPLQKPKQDRYALRTSPQWLGPQVE   | 358 |
| SmPAL1 | VIRTATKMIEREINSVNDNPLIDVSRNKALHGGNFQGTPIGVSMDNARLAIASIGKLLFA  | 420 |
| SmPAL2 | VIRAATKSIEREINSVNDNPLIDVSRNKALHGGNFQGTPIGVSMONTRLAIASIGKLMFA  | 418 |
| SmPAL3 | MIRTATKMIEREINSVNDNPLIDVSRNKALHGGNFQGTPIGVSMONSRLAIASIGKLMFA  | 418 |
| SmPAL1 | QFSELVNDLYNGLPSNLGGGRNPSLDYGFKGSEIAMASYCSELQFLANPVTNHVQSAEQ   | 480 |
| SmPAL2 | QFSELVNDLYNGLPSNLGGGRDPSLDYGFKGAEIAMAAYCSELQFMANPVTSHVQSAEQ   | 478 |
| SmPAL3 | QFSELVNDLYNGLPSNLGGGRNPSLDYGFKGAEIAMAASYCSELQFLANPVTNHVQSAEQ  | 478 |
| SmPAL1 | HNQDVNSLGLISSRKTVEALDILKLSSTYLVALCQAVDLRHLEENLKHAVKNVTSQVAK   | 540 |
| SmPAL2 | HNQDVNSLGLISARKTAEAVEILKLSSTYLVALCQAMDRLHLEENLKHAVKNVTSQVAK   | 538 |
| SmPAL3 | HNQDVNSLGLISSRKTVEALDILKLSSTFLVALCQVIDLRHLEENLRLAVKNVTSQVAK   | 538 |
| SmPAL1 | RTLTMGVNGELHPSRFCEKDLIRVVDREYVFAYIDDPSSATYPLMQKLRQVLVDHALKNG  | 600 |
| SmPAL2 | KVLTMGQNGELHPSRFSEKELKVVREHVFYIDDPSSSNYPLMQRLRQVLVDHALANG     | 598 |
| SmPAL3 | RTLTMGANGELHPSRFCEKDLIRVVDREYVFAYVDDPCLATYPLMQKLRQVLVDHALKNG  | 598 |
| SmPAL1 | DLEKNASTSIFQKIEAFEEELKALLPKEVGSARMALESQSPTVANRIAECRSYPLYKFIR  | 660 |
| SmPAL2 | DAEKEASTSIFLKIGAFEEELKAMLPKEVEAARVEVEKGRAAIESRIKGRSRYPLYRFVR  | 658 |
| SmPAL3 | DGEKNPSTSIFQKIEAFEALEVLPLKEVEGARSALAEAGNPAIGNRIKCRSYPLYKFVR   | 658 |
| SmPAL1 | EQLGAGFLTGEKAVSPGEECEKVFTALSNGLIIDPLECLQGWNGQPLPI             | 710 |
| SmPAL2 | EEAGTGFLTGEKARSPGEEFDKVFRCMCEGLIDPLMDCLREWNGAPLPI             | 708 |
| SmPAL3 | EELGAFLTGEKATSPGEEGEKVFTALSNGLLVDPPLKCLSWNGEPLPI              | 708 |

expression levels (Fig. 5). *SmPAL1*, the most abundant among three *SmPALs* in all of the tissues analyzed, showed the highest expression in roots, followed by leaves, stems and flowers. *SmPAL2* was predominately expressed in stems and flowers. Its expression levels in roots and leaves were very low. *SmPAL3* had the highest expression in leaves, less in roots, stems and flowers. The results suggest *SmPAL1* and *SmPAL3* can function in all of the tissues analyzed, whereas *SmPAL2* appears to play roles mainly in stems and flowers under normal conditions.

#### Responses of *SmPAL* genes to PEG-6000 and MeJA treatments

The presence of many environmental signal-responsive *cis*-elements in the promoters of *SmPAL* genes suggests that

*SmPAL* genes might be involved in plant response to stress. To test this hypothesis, we analyzed the expression level of *SmPAL* genes in roots of plantlets treated with PEG-6000 and MeJA using qRT-PCR. The results showed that all three *SmPALs* were regulated by drought and MeJA treatments, although the time and degree of reaction differed from one another (Fig. 6). Under drought conditions, *SmPAL1* and *SmPAL3* were up-regulated to about three-folds after being treated for 24 and 36 h, respectively (Fig. 6). The up-regulated level of *SmPAL2* after 36 h drought treatment was the most significant, which reached to about ninefolds (Fig. 6). Similar results were also observed for *SmPALs* in roots of plantlets treated with MeJA for 24 and 36 h (Fig. 6). However, down-regulation was found for all *SmPALs* after 12 h MeJA treatment and *SmPAL1* was up-regulated against after 48 h treatment in



**Fig. 4** Joined phylogenetic tree of PALs from *S. miltiorrhiza* and other organisms. The deduced full-length amino acid sequences were aligned using ClustalW version 1.83 and the phylogenetic tree was constructed using MEGA 4.0 by the neighbor-joining (NJ) method with 100 bootstrap replicates. PALs from fungus, gymnosperms, monocots and dicots are indicated with backgrounds in purple, blue, red, and green, respectively. SmPALs are highlighted with yellow. Ar *Agastache rugosa*, At *Arabidopsis thaliana*, Bo *Bambusa oldhamii*, Ca *Coffea arabica*, Cc *Coffea canephora*, Cd *Cistanche deserticola*,

Ci *Citrus limon*, Cr *Catharanthus roseus*, Cs *Camellia sinensis*, Dc *Daucus carota*, Dl *Digitalis lanata*, Ep *Euphorbia pulcherrima*, Gb *Ginkgo biloba*, In *Ipomoea nil*, Jc *Jatropha curcas*, Me *Manihot esculenta*, Mo *Melissa officinalis*, Na *Nicotiana attenuata*, Nt *Nicotiana tabacum*, Os *Oryza sativa*, Pf *Perilla frutescens*, Pp *Pinus pinaster*, Pt *Populus trichocarpa*, Pta *Pinus taeda*, Rg *Rehmannia glutinosa*, Rt *Rhodospiridium toruloides*, Sb *Scutellaria baicalensis*, Sv *Scutellaria viscidula*, Ta *Triticum aestivum*, Vv *Vitis vinifera*, Zm *Zea mays*. (Color figure online)

**Table 2** Number of various *cis*-acting regulatory elements existing in the promoters of *SmPAL* genes

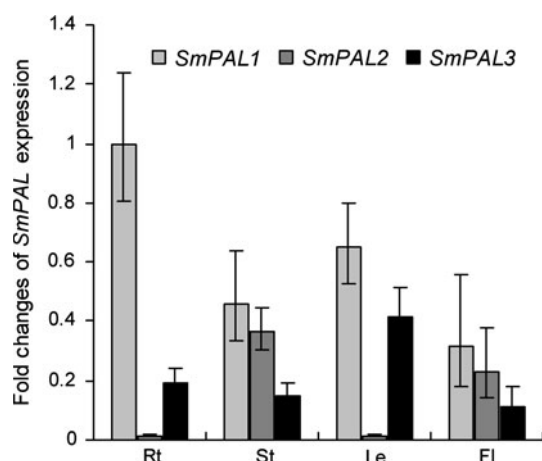
| <i>Cis</i> -elements           | <i>SmPAL1</i> | <i>SmPAL2</i> | <i>SmPAL3</i> |
|--------------------------------|---------------|---------------|---------------|
| Box L (YCYACWACC)              | 1             | 1             | 1             |
| AC element (CYACWACC)          | 1             | 1             | 1             |
| MYB2 recognition site (YAACKG) | 1             | 1             | 1             |
| E-box (CANNTG)                 | 1             | 4             | 1             |
| AG motif (AGATCCAA)            | 1             | 0             | 0             |
| Box A (CCGTCC)                 | 0             | 0             | 1             |
| Box P (YTYMMCMAMCMC)           | 0             | 0             | 0             |

addition to the up-regulation after 24 h treatment, which were different from the results observed for drought treatment (Fig. 6). These results suggest that all of three *SmPALs* are drought- and MeJA- responsive. Among them,

*SmPAL1*, which was induced significantly in plantlets treated for 24 h, showed the quickest response to both drought and MeJA treatments, whereas *SmPAL2*, which was induced to about nine and 18-folds after 36 h treatment, had the most significant response to these treatments. Considering the low expression of *SmPAL2* in roots of plants cultivated under normal conditions, the significant change of *SmPAL2* transcripts could be important for plant response to environmental stress.

## Discussion

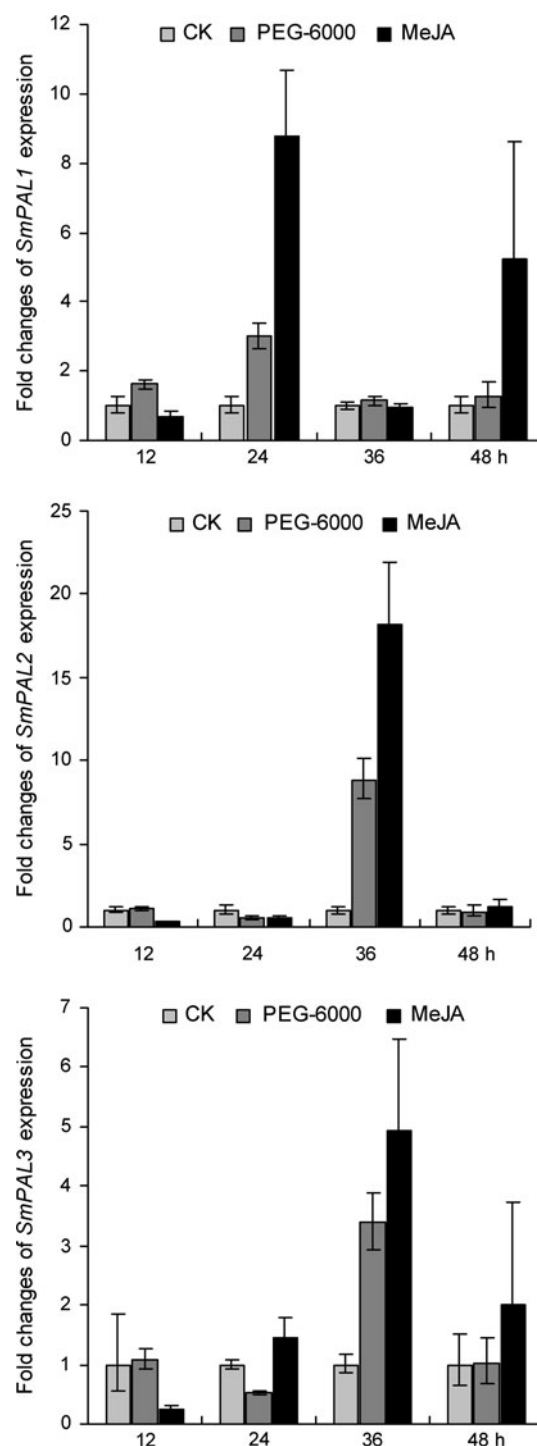
PAL is a significant key enzyme in the biosynthesis of many useful secondary metabolites. In this study, we successfully performed a genome-wide search of *SmPAL* genes by BLAST analysis of the current assembly of *S.*



**Fig. 5** Tissue-specific expression of *SmPALs* in *S. miltiorrhiza*. Fold changes of *SmPALs* in *Rt* roots, *St* stems, *Le* leaves and *Fl* flowers of two-year-old *S. miltiorrhiza* plants are shown. The expression level of *SmPAL1* in roots was arbitrarily set to 1

*miltiorrhiza* genome (Chen et al. unpublished data). A total of three *SmPAL* genes were identified. Using PCR technology, we cloned the full-length *SmPAL3* cDNA and the 5'-region of *SmPAL2* cDNA that were unknown previously. The capability of the identified genes in encoding PAL in *S. miltiorrhiza* was confirmed by high sequence similarity between *SmPALs* and known PALs from *A. thaliana* [8], *M. officinalis* [26], *A. rugosa* [27], *P. frutescens*, *S. baicalensis* [11], *S. viscidula* [28], and so on. The existence of a small multigene family in *S. miltiorrhiza* is consistent with the results from other plant species with whole genome sequence available [8–10, 12].

Phylogenetic analysis of *SmPALs* and PALs from other organisms showed that PALs from fungus, gymnosperms, monocots and dicots clustered to four distinct clades and PALs from dicots might be divided into four sub-clades (Fig. 4). The identified three *SmPALs* fell into two different sub-clades of dicot PALs with *SmPAL1* and *SmPAL3* in a sub-clade whereas *SmPAL2* in the other one. Consistently, the identity of *SmPAL1* and *SmPAL3* at both the nucleotide sequence level and the amino acid sequence level is higher than that of *SmPAL1* and *SmPAL2* and of *SmPAL2* and *SmPAL3*. It indicates the role of *SmPAL2* is different from *SmPAL1* and *SmPAL3* in some aspects in *S. miltiorrhiza*. Since *SmPAL3* clustered in a sub-clade with MoPAL and *SmPAL1* that were involved in the biosynthesis of rosmarinic acid [24, 26] and CdPAL that was associated with the biosynthesis of phenolic compounds [29], it is very likely that *SmPAL3* is also involved in rosmarinic acid biosynthesis in *S. miltiorrhiza*. However, it is currently unknown whether *SmPAL2* is involved in rosmarinic acid biosynthesis because there is no information about the role of PALs clustered with *SmPAL2* in the biosynthesis of rosmarinic acid.



**Fig. 6** Stress responses of *SmPALs* in *S. miltiorrhiza*. Fold changes of *SmPALs* in roots of *S. miltiorrhiza* plantlets treated with MeJA or PEG-6000 for 12, 24, 36 and 48 h. The level of transcripts in roots without treatment was arbitrarily set to 1

It has been shown that different member of the *PAL* gene family in a plant may have distinguishable biochemical, molecular and catalytic properties [38] and involved in the production of different products under



specific conditions [39]. In coffee bean, *CcPAL1* and *CcPAL3* are associated with the accumulation of chlorogenic acids (CGA), whereas *CcPAL2* may contribute more significantly to flavonoid accumulation [13]. Similar results were also observed for two *P. tremuloides* *PALs*, of which *PtPAL1* was involved in condensed tannin metabolism and *PtPAL2* was associated with monolignol biosynthesis [14]. Consistently, differential expression patterns were observed for three *SmpALs*, although they were expressed in all of the tissues analyzed (Fig. 5). The transcripts of *SmpAL1* and *SmpAL3* showed high levels in roots and leaves, whereas *SmpAL2* were predominately expressed in stems and flowers. It indicates that specific products of the phenylpropanoid pathway are probably synthesized through a metabolic channel organized by specific isoenzymes. However, it has also been shown that different *PAL* genes in a plant may coordinately function together in the production of a specific product. For instance, *A. thaliana* *AtPAL1*, *AtPAL2* and *AtPAL4* were associated with lignin biosynthesis, and *AtPAL1* and *AtPAL2* were also involved in the biosynthesis of stress-induced flavonoids [8, 40, 41]. In *S. baicalensis*, all of three *SbPAL* genes were associated with stress-induced flavonoid synthesis [11]. In this study, *SmpAL1* and *SmpAL3* were found to cluster with *SbPAL2* and *SbPAL3* in a sub-clade of dicot *PALs*, and *SmpAL2* clustered with *SbPAL1* in the other sub-clade (Fig. 4). It indicates that all three *SmpALs* may function redundantly as *SbPALs* in the biosynthesis of flavonoids in *S. miltiorrhiza*.

Analysis of the 5'-flanking regions of *SmpAL* genes showed the existence of several conserved *cis*-elements, such as elicitor- or light-associated box L and AC element (CYCACCCWACC) [33], drought-related MYB2 recognition site [35] and MeJA treatment-responsive E-box [36], suggesting that *SmpALs* could be functionally redundant in response to some environment stresses in *S. miltiorrhiza* (Table 2). Consistently, the expression of *SmpALs* was all induced in roots of plantlets treated with PEG-6000 and MeJA (Fig. 6). On the other hand, difference was observed for three *SmpALs* in the time and degree of response after PEG-6000 and MeJA treatments. Similar results were previously found for three *PALs* from *S. baicalensis* [11]. All three *SbPALs* responded to MeJA treatment, but their responses were different. *SbPAL1* showed the highest induction by 100  $\mu$ M MeJA, whereas *SbPAL2* and *SbPAL3* were induced to the highest degree by 200  $\mu$ M MeJA. In the tissues treated with 100  $\mu$ M MeJA, *SbPAL1* and *SbPAL2* showed the highest induction in 48 h, while *SbPAL3* reached to the highest point after 12 h treatment [11]. Because many of the downstream products of the phenylpropanoid pathway, such as lignins, flavonoids, coumarins, isoflavonoids, furanocoumarins and norlignans, are cell wall constituents, pigments, UV protectants and

plant defense compounds [42–44], the subtle regulation of *PAL* gene expression could be very important for plants to adapt to the stressful environments.

Taken together, our results showed the existence of at least three *SmpALs* in *S. miltiorrhiza*. These *SmpALs* may function redundantly in the biosynthesis of some metabolites, but the roles of each *SmpAL* maybe different in the biosynthesis of other metabolites, suggesting the complexity of *SmpAL*-associated metabolic network in *S. miltiorrhiza*. Further analyzing the physiological functions of *SmpALs* using transgenic approaches will definitely shed light on elucidating the metabolic network of secondary metabolites in *S. miltiorrhiza*.

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